Sparse MoE from K2 to K3

Deep Dive into Kimi K3 — Chapter 3

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Foundation Books Project

The one idea of this chapter

Sparse experts grow stored capacity without evaluating every expert for every token — and the result depends on *selection*, *combination*, and *where* the chosen work runs.

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Goal: route ten tokens into twenty auditable records and back to ten outputs, then read K2's sparsity evidence and K3's latent routing through ledgers that each keep their own unit.

Roadmap

The typed sparse layer

Routing and the route trace

Capacity and balance

K2 evidence and the K3 change

Placement and diagnosis

The typed sparse layer

Shared always, routed by selection

$$\mathrm{MoE}(x_t) = \sum_{j=1}^{N_s} S_j(x_t) + \eta \sum_{i \in \mathcal{T}_t} p_{t,i} E_i(x_t) \in \mathbb{R}^d$$

- Toy: $d = 8$, $E = 4$, $k = 2$, $N_s = 1$; ten origins $v_1..v_4, \tau_1..\tau_6$.

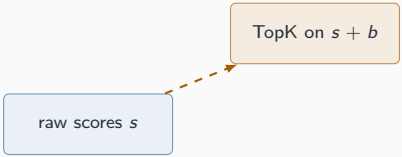
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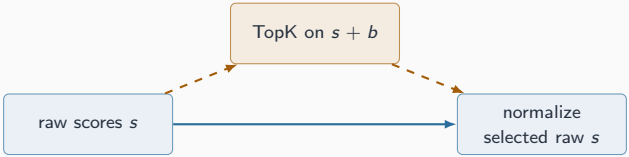
- Toy: $d = 8$, $E = 4$, $k = 2$, $N_s = 1$; ten origins $v_1..v_4, \tau_1..\tau_6$.
- Five new contracts: membership, weighting, identity, load, placement — k/E alone answers none of them.

Routing and the route trace

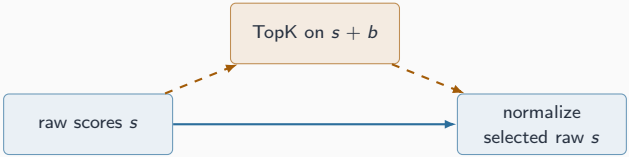
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Row 7: experts $(4, 1)$, weights $(0.55154, 0.44846)$. Leaking b into the weights keeps unit sums and still changes the mixture $(5.515 \to 5.070)$.

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A wrong-origin return passes shape and count checks — only its tag reveals the fault.

Capacity and balance

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- At $c = 1.2$, $\kappa = 6$: one overflow at Expert 1, five padded slots elsewhere — no c removes both.
- Loss-free sign update: $b \to (0.15, -0.15, 0.05, 0.15)$, for the *next* decision only, never in the weights.

K2 evidence and the K3 change

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K3's $\approx 2.5\times$ scaling gain is attributed collectively to architecture, data, and recipes — not to sparse MoE alone.

Placement and diagnosis

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Expert, rank, and execution balance are different levels; diagnose cheap contracts first, then loads, then placement, then time.